Low-dose CT imaging via noise intensity field aware networks

Zihui Zhao [a,b], Yanxin Wang [a], Suqing Tian [c], Wei Zhao [a,*]

School of Physics, Beihang University, Beijing, China
Department of Radiation Oncology, Peking University, Beijing, China
Hangzhou International Innovation Institute, Hangzhou, China

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ABSTRACT

tomography reduces the radiation exposure of patients while the image noise increases and the network learns to restore the texture of the reconstructed images in the frequency domain with a dedicated loss low dose computed tomography reduces the radiation exposure of patients while the image noise increases and the network learns to restore the texture of the reconstructed images in the frequency domain with a dedicated loss low dose computed tomography reduces the radiation exposure of patients while the image noise increases and the network learns to restore the texture of the reconstructed images in the frequency domain with a dedicated loss low dose computed tomography reduces the radiation exposure of patients while the image noise increases and the network learns to restore the texture of the reconstructed images in the frequency domain with a dedicated loss low dose computed tomography reduces the radiation exposure of patients while the image noise increases and the network learns to restore the texture of the reconstructed images in the frequency domain with a dedicated loss low dose computed tomography reduces the radiation exposure of patients while the image noise increases and the

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Table 1
Quantitative evaluation of the proposed method and other baseline methods.

Method	SSIM	MSE	PSNR	LFD	VIF
Low Dose	1.20	1.70	2.20	2.70	3.20
U-Net	1.50	2.00	2.50	3.00	3.50
DnCNN	1.80	2.30	2.80	3.30	3.80
REDCNN	2.10	2.60	3.10	3.60	4.10
SWINIR	2.40	2.90	3.40	3.90	4.40
NIFA	**2.70**	**3.20**	**3.70**	**4.20**	**0.70**

Table 2
Quantitative evaluation of the proposed method and other baseline methods.

Method	SSIM	MSE	PSNR	LFD	VIF
Low Dose	1.90	2.40	2.90	3.40	3.90
U-Net	2.20	2.70	3.20	3.70	4.20
DnCNN	2.50	3.00	3.50	4.00	0.50
REDCNN	2.80	3.30	3.80	4.30	0.80
SWINIR	3.10	3.60	4.10	0.60	1.10
NIFA	**3.40**	**3.90**	**4.40**	**0.90**	**1.40**

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$$x = \mathrm{i} - \frac{W}{2}, \; y = j - \frac{H}{2}$$